

1. Introduction

1.1 Purpose:

The purpose of this Software Requirements Specification is to define the requirements for SplitsFromVideo, as well as provide a structured overview of the specifications of the project. This document will be the main reference point while implementing and designing the software itself. The audience for this document is the developers of the project, the project's faculty advisor, and the professor managing senior design.

1.2 Scope:

SplitsFromVideo aims to allow for speedrunners to extract split information from a speedrun YouTube video. In a typical speedrun YouTube video, a LiveSplit timer is present on screen that shows information about each split of the speedrun. Currently, if someone wants to extract the information from this on-screen timer, they must manually skip through the video looking for the last video frame of each split. This is very tedious and requires a lot of manual seeking. This method is also prone to human error, which is undesirable in a speedrun setting, as speedrunners strive to be precise down to the decimal for timing. Furthermore, there may be a loss of information in this method based on how much information the on-screen timer shows, such as how many decimal places are present.

The primary functionality of SplitsFromVideo will be to automatically find the length of each split in the input speedrun YouTube video and return them to the user in the desired format(s). The software will be able to identify on exactly which frame a given split ends and extract extra frame information from the video that is not present in the on-screen timer. This will allow extra precision in split lengths without the substantial extra manual effort required for such precision. Additionally, the software will be able to output the splits in several formats, including plaintext, LiveSplit splits (.lss), and csv.

1.3 Definitions & Acronyms:

- Speedrunning - the act of playing a video game with the goal of completing it as fast as possible.
- Splits - specific segment times used to divide a full speedrun into smaller parts.
- LiveSplit - An easy-to-use timer/splits program for speedrunners. It can show delta time comparisons, handle timer load removal, and provide additional information about the runner's current and past speedruns.
- speedrun.com (SRC) - a website containing speedrun leaderboards for different games and their categories.
- Personal best (PB) - a speedrunner's best time they've achieved for a specific speedrun.
- World record (WR) - the best time ever achieved for a specific speedrun.
- Gold split - A speedrunner's best time for a specific split in a speedrun.

- LiveSplit comparison - a given set of splits that LiveSplit is able to compare to in order to generate deltas.
- LiveSplit splits file (.lss) - the XML file format used by LiveSplit to store the split information for a speedrun. This file exists per speedrun category.

1.4 References:

- [1] IEEE Computer Society, "IEEE Recommended Practice for Software Requirements Specifications," IEEE Std 830-1998, 1998.
- [2] OpenCV Development Team, "OpenCV Reference Manual," 2024. [Online]. Available: <https://docs.opencv.org/>
- [3] PaddlePaddle Authors, "PaddleOCR Documentation," 2024. [Online]. Available: <https://paddlepaddle.github.io/PaddleOCR/>
- [4] yt-dlp Contributors, "yt-dlp Documentation," 2024. [Online]. Available: <https://github.com/yt-dlp/yt-dlp>

1.5 Overview:

The remainder of this document is organized into two main sections. Section 2 provides an overall description of the product, including its perspective, user functions, user characteristics, and operating constraints. Section 3 details the specific software requirements, including external interfaces, functional requirements, performance metrics, and system quality attributes.

2. Overall Description

2.1 Product Perspective:

While SplitsFromVideo is a standalone app, it will use several third-party libraries to extract video frames and recognize digits on screen. Python will be used to handle the backend logic of the system. The Python libraries yt-dlp and cap-from-youtube will be used for video acquisition. These libraries will stream the video directly from YouTube based on the URL. OpenCV will be used for on-screen timer isolation. PaddleOCR will be used to obtain the digit information from the on-screen timer. Pywebview will be used alongside HTML and JavaScript to make the GUI of the program.

2.2 Product Functions:

- The software shall take a YouTube video link as input.
- The software shall provide a video player view of the video.
- The software shall prompt the user to select where the run ends in the video.

- The software shall provide an interface to click and drag to select where the timer is on-screen using the video player interface.
- The software shall crop each frame of the video obtained to the bounding box selected.
- The software shall parse timer data using PaddleOCR to determine where to seek to.
- The software shall find the end of each split and keep track of how long it is.
- The software shall output the extracted split information from the video in plaintext format.
- The software shall output the extracted split information from the video in LiveSplit layout (.lss) format.
- The software shall output the extracted split information from the video in comma separated variables (.csv) format.

2.3 User Classes:

Due to SplitsFromVideo being a standalone application, there is no standard role-based system for accounts. However, it is designed to be useful to different types of users within the speedrun community.

- Speedrunner - the primary user of the application. Has general speedrunning knowledge/experience and an understanding of the applications of the gathered information from the program. This user will be able to use the provided information to create LiveSplit comparisons.
- Speedrun data analyst - the secondary user of the application. This user is expected to have general speedrunning knowledge and understanding of the data applications. This user will be able to use the provided information to analyze trends, create predictions, and document the progress of a runner. Most often, this user will use the data in speedrun documentary video creation.

2.4 Operating Environment:

SplitsFromVideo is designed to run as a desktop executable program. Due to the nature of processing video frames from an online source (YouTube), as well as using machine learning libraries for text recognition, there are specific hardware and network requirements.

Hardware requirements:

- Processor - Multi-core processor required for OpenCV, PaddleOCR, and multi-threading support.
- Memory - at least 4 GB of RAM required for use.
- Storage - at least 500 MB of storage for the executable and dependencies included.

Operating System requirements:

- Windows - Windows 10 or 11

- macOS - macOS 10.15 or higher
- Linux - standard modern distributions

Network requirements:

- Internet connection - required for use of yt-dlp connecting to YouTube servers.
- Port Access - Port 443 open to allow HTTPS requests

2.5 Assumptions:

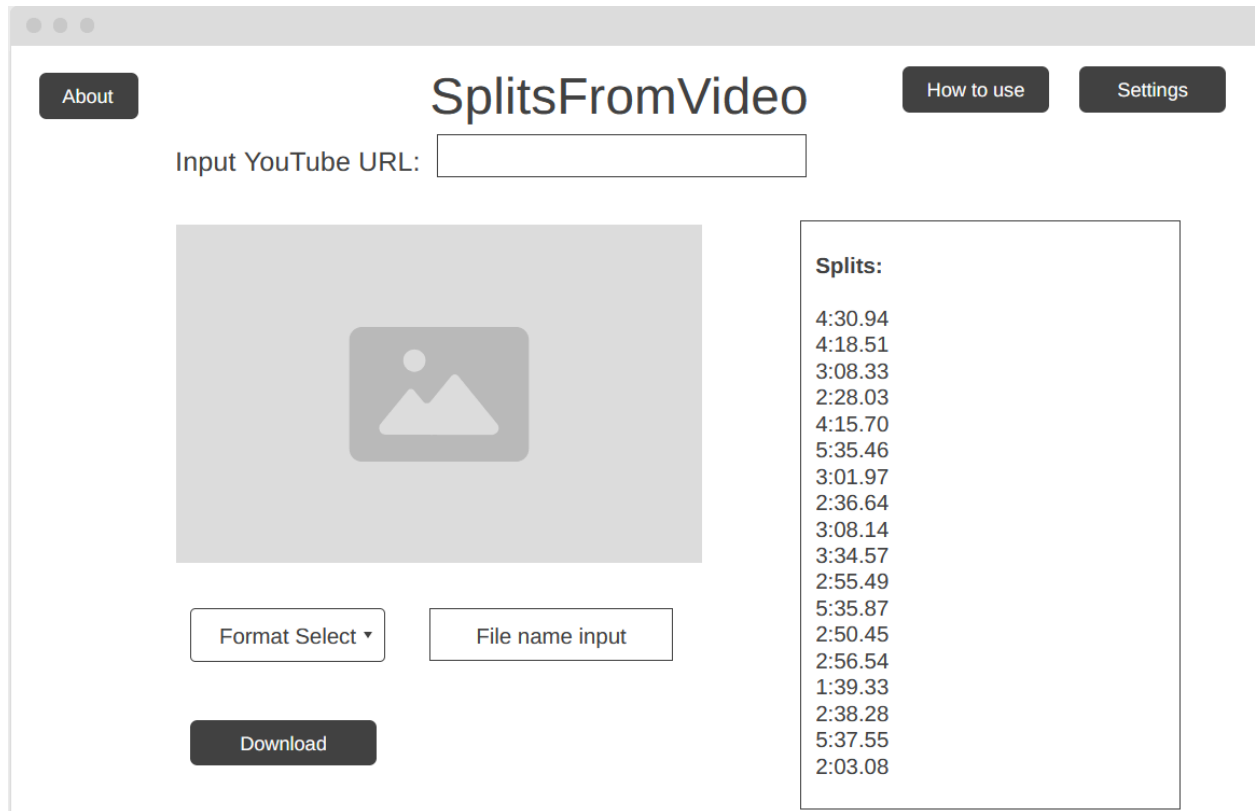
- Assumption 1: Target YouTube videos contain a visible, unobstructed on-screen timer in a fixed screen position throughout the run.
- Dependency 1: Video intake relies on the functionality and availability of YouTube's servers as well as the yt-dlp library.
- Dependency 2: The client must maintain an active internet connection which can stream video content at 720p or higher.

3. Specific Requirements

3.1 External Interface Requirements:

User Interfaces (UI):

- Header Navigation (Top): The software shall display the application title SplitsFromVideo, as well as about, how to use, and settings buttons.
- URL Input (Top Center): The software shall have a text input field for the user to provide the desired video link.
- Video player (Left Column): The software shall display the video player on the left column, where users will define the bounding box for OCR extraction.
- Data Review Panel (Right Column): The software shall display a vertical, scrollable splits list in the right column. This fills dynamically as the video is analyzed and the splits are extracted.
- Export Controls (Bottom Left): The software shall provide a dropdown to select the desired output format, a text input field to name the file, and a download button.



(Wireframe mockup of the program's user interface)

Hardware Interfaces:

This system does not interface with any meaningful external peripherals.

Software Interfaces:

- SI-01 - Pywebview: the software shall interface with Pywebview to help render and use HTML and JavaScript to shape the user interface on the native operating system.
- SI-02 - yt-dlp: the software shall establish a video stream using the yt-dlp Python library.
- SI-03 - OpenCV: the software shall use OpenCV to process the data from the video stream.
- SI-04 - PaddleOCR: the software shall give PaddleOCR the processed OpenCV data to extract the timer text information into a usable format.

Communications Interfaces:

- CI-01 - HTTPS: the software shall use HTTPS GET requests with yt-dlp to establish the video stream.
- CI-02 - TCP/IP: the software will use standard network protocols to communicate with YouTube.

3.2 Functional Requirements:

Input and User Interface Functions:

- FR-01 - The software shall accept a valid YouTube video URL as text input.
- FR-02 - The software shall show a video player view of the inputted YouTube video in the user interface.
- FR-03 - The software shall provide a click and drag interface over the video player for the user to define a bounding box.
- FR-04 - The software shall store the screen coordinates of the user-defined bounding box to isolate the on-screen timer.
- FR-05 - The software shall prompt the user to input or select the timestamp where the speedrun ends.

Processing and Data Extraction Functions:

- FR-06 - The software shall crop every processed video frame to match the exact coordinates of the defined bounding box.
- FR-07 - The software shall process the cropped frames using the PaddleOCR engine to extract the timer digits as text data.
- FR-08 - The software shall use the parsed OCR time data to determine the endpoints of individual splits.
- FR-09 - The software shall calculate and store the total duration of each split.

Output and Export Functions:

- FR-10 - The software shall export the extracted split data into a .csv file format.
- FR-11 - The software shall export the extracted split data into a LiveSplit .lss file format.

- FR-12 - The software shall export the extracted split data into a standard plaintext (.txt) file format.

3.3 Performance Requirements:

- PR-01 - The software shall initialize and render the video stream from a valid YouTube URL within 5 seconds of user submission
- PR-02 - The software shall process the cropped video frames through the PaddleOCR engine at a minimum extraction rate of 5 frames per second.
- PR-03 - The software shall generate and save the export file within 2 seconds of the user clicking the download button.
- PR-04 - The software's OCR implementation shall achieve a minimum digit recognition accuracy of 95% when processing unobstructed, high-contrast timer displays at a source resolution of 720p or greater.

3.4 Software System Attributes:

Security:

- The software shall execute entirely within local user space without requiring administrative or elevated operating system privileges.
- The software shall not collect, store, or transmit personally identifiable information.

Reliability:

- The software shall handle network connection issues and interruptions properly without causing unexpected software terminations.
- In the event of an unreadable frame or OCR extraction error on a specific image crop, the software shall log the frame exception.

Maintainability:

- The software shall automatically check for updates to third-party libraries.
- The software shall automatically check for updates to the software version.

Portability:

- The software shall be written with cross-platform compatible Python libraries.
- The graphical user interface will use the native web engine of the operating system to render the Pywebview user interface.